

LedgerFactor

Confidential invoice financing on Canton. A supplier discounts a buyer-approved invoice to a financier, and the financier's margin is hidden from the buyer **by the protocol**. The ledger **structurally** guarantees one invoice can never be financed twice.

LIVE AND DAML-SCRIPT PROVEN

DAML 2.10.4 ON CANTON

A supplier waits 60 days. A financier will pay today, for a price.

Invoice factoring is a **4 trillion dollar** market that runs on one commercial secret: the discount rate the financier charges. Put that trade on a shared ledger and two things break.

BREAK 1 · PRICE LEAK

The buyer sees what the financier earns. That is **negotiating leverage against the supplier** on every future invoice, and it is the reason confidential factoring stays off-chain.

BREAK 2 · DOUBLE PLEDGE

The same receivable financed twice to two lenders. Classic factoring fraud. Off-chain it is caught by **registries and trust**, after the fact.

THE BIND

Encrypt it and you lose the shared truth that prevents the double pledge. Publish it and you lose the price. **You need both at once.**

Three guarantees, enforced by the ledger and not by the UI

SELECTIVE DISCLOSURE

The rate lives on **FinancingProposal** / **FinancingOffer**, whose only stakeholders are the financier and the supplier. The buyer is never a stakeholder, so the margin **never reaches the buyer's party view**. Not hidden in a view. Not there.

SINGLE EXERCISE

The Invoice is one authoritative contract. **AcceptFinancing is a consuming choice**, so a second financing attempt on the same invoice is rejected by the ledger itself. Anti-double-pledge is structural, not a check we remembered to write.

ATOMIC DVP

Assigning the receivable and paying the supplier the discounted cash **settle in a single transaction**. No half-financed state exists.

On a public chain the sealed bid leaks through logs before the reveal, and the margin is a matter of public record. **Canton's party-scoped disclosure is the product**, not a feature of it.

SUPPLIER

Northwind Supply

Issues & discounts the receivable

Sup7f0c0f7db...f8eb

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMED listed to financier

FINANCING OFFER · CONFIDENTIAL

MARGIN / DISCOUNT

1.97%

ADVANCE TO
\$98,030
disclosed: financier + supplier

2 contracts

BUYER

Globex Corp

confirms the payable

Buy7f0c0f7db...f8eb

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMED listed to financier

FINANCING TERMS

Withheld by the Canton ledger from this participant. A margin exists on a contract this party is not a stakeholder of.

Withheld by the Canton ledger from this participant. A margin exists on a contract this party is not a stakeholder of.

FINANCIER

Meridian Capital

underwrites & funds

Fin7f0c0f7db...f8eb

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMED listed to financier

AI UNDERWRITING · Q3 PALLET DELIVERY

89/100 A

APPROVE

recommended discount

1.97%

RELIABILITY

CONCENTRATION

DILUTION

SIZE

FINANCING OFFER · CONFIDENTIAL

MARGIN / DISCOUNT

1.97%

ADVANCE TO SUPPLIER

\$98,030

FINANCIER SPREAD

\$1,970

disclosed: financier + supplier

AUDITOR

Regulator

04 / 09

observes the audit trail

Aud7f0c0f7db...f8eb

FINANCING TERMS

Withheld by the Canton ledger from this participant. A margin exists on a contract this party is not a stakeholder of.

Withheld by the Canton ledger from this participant. A margin exists on a contract this party is not a stakeholder of.

0 contracts in this party's view

THE MONEY-SHOT · FOUR LIVE PARTY VIEWS

The 1.97% margin reaches two of four parties

Supplier and financier see the rate. Buyer and auditor get a redaction bar — their party view was never sent the contract.

No financier can see a rival's bid

As a financier, **only your own envelope opens**. Rivals stay wax-sealed. Proven by **testSealedBidAuction**.

VIEW LEDGER AS

YOU HOLD ONE KEY, AS

Meridian Capital

FINANCIER

single key · only your own envelope opens; rivals stay wax-sealed

AUCTIONEER

Supplier

FINANCIER

Meridian Capital

FINANCIER

Apex Credit

FINANCIER

Cobalt Partners

OBSERVER

Buyer

OBSERVER

Auditor

MERIDIAN CAPITAL · AGGRESSIVE BIDDER

You see **only your own** envelope. Rivals stay **wax-sealed** - the ledger never sent you their rate.

SEALED LOT · BUYER-APPROVED RECEIVABLE

FACE VALUE

CONFIRMED

Q3 pallet delivery

\$100,000

01

Meridian Capital

DISCOUNT RATE · DISCLOSED TO YOU

2.30%

ADVANCE TO SUPPLIER

\$97,700

FINANCIER SPREAD

\$2,300

THIN MARGINS · PRIME SMALL-TICKET

02

Apex Credit

SEALED BID

BALANCED BOOK · MID-MARKET

03

Cobalt Partners

SEALED BID

DEEP CAPACITY · LARGE-TICKET

05 / 09 · sealed bids would leak via public-chain logs before the reveal

An underwriting engine, not a chatbot

The agent underwrites the receivable over **only the data the financier is entitled to see**. That is the point: the AI inherits the ledger's disclosure boundary, it does not reach around it.

DETERMINISTIC CORE

Scores buyer payment reliability, portfolio concentration and dilution risk into a **0-100 credit score**, a risk band and a recommended discount rate. **10/10 unit tests**.

HUMAN IN THE LOOP

The agent recommends. **The financier reviews and signs** the resulting on-ledger offer. The rate the demo executes is the rate the agent actually recommended.

NO NETWORK DEPENDENCY

An LLM turns the structured rationale into a short memo. With no API key it **falls back to a deterministic memo**, so the demo never depends on a network call.

Daml Script tests, each querying the ledger as its own party

```
$ daml test
daml/Tests.daml:setupParties: ok, 0 active contracts, 0 transactions.
daml/Tests.daml:testSelectiveDisclosure: ok, 3 active contracts, 4 transactions.
daml/Tests.daml:testNoDoubleFinancing: ok, 6 active contracts, 11 transactions.
daml/Tests.daml:testHappyPathSettlement: ok, 4 active contracts, 7 transactions.
daml/Tests.daml:testSealedBidAuction: ok, 6 active contracts, 10 transactions.
```

DISCLOSURE

The buyer queries for FinancingOffer and gets **nothing**. The financier and supplier see the margin.

DOUBLE FINANCE

Two offers on one invoice. The first AcceptFinancing consumes it, the second is **rejected by the ledger**.

SETTLEMENT

Atomic DvP: supplier paid **97,000 on a 100,000 invoice**, original receivable gone, auditor sees face value only.

What is real, and what is not

The privacy and the atomicity are real and proven. Two things are deliberately scoped, and we would rather say so than be found out.

MOCK CASH

Cash is an owner-signed bearer token standing in for a token-standard holding. The DvP **atomicity is real**; the cash model is not production grade. Real token-standard DvP under **CIP-0056** is the stretch goal.

NOTIFIED FACTORING

Listing an invoice reveals the financier's identity to the buyer. The sensitive figure, **the pricing, stays hidden**. Undisclosed factoring would use Canton explicit disclosure.

Everything else on these slides is running at ledgerfactor.unitynodes.com right now and reproducible from the public repo.

Built, deployed, and provable in one command

SHIPPED

Daml model with both money-shots proven by Daml Script · AI credit-scoring agent, unit-tested · live JSON API gateway with party-scoped views · four-role frontend · **Veild sealed-bid auction** · persistent deployment behind Caddy and Cloudflare

NEXT

Real token-standard DvP settlement under CIP-0056 · runtime compliance citation in the auditor view

LIVE DEMO

ledgerfactor.unitynodes.com

SOURCE, `APACHE-2.0`

`github.com/UnityNodes/ledgerfactor`